

PROGRAM ANNOUNCEMENT: PA-20-185 — NIH RESEARCH PROJECT GRANT (R01)

Project Title: Next-Generation Synthetic Nanocarriers for Target-Specific BBB Transcytosis and Glial Remodeling in Neurodegeneration

Principal Investigator: Dr. Elena M. Vance, Ph.D. (Synapse Institute)

Co-Investigators: Dr. Aris Thorne (Apex Labs), Dr. Marcus Lin (Vertex)

Lead Institution: Synapse Institute for Biomedical Research

FOA: PA-20-185 (NINDS)

Period: 09/2026 – 08/2031

Direct Costs: \$1.25M (5 Yrs)

1. A. SIGNIFICANCE**1.1 A.1 Scientific Premise and Clinical Unmet Need**

Neurodegenerative disorders, including Alzheimer's disease (AD) and Amyotrophic Lateral Sclerosis (ALS), represent a critical public health crisis affecting over 6.5 million Americans, with projected care costs exceeding \$1.1 trillion by 2050. A major rate-limiting barrier in disease-modifying pharmacotherapy is the **Blood-Brain Barrier (BBB)**, a highly restrictive microvascular endothelium that excludes >98% of small-molecule drugs and >99% of macromolecular biologics, antibodies, and nucleic acid therapeutics. Existing delivery vehicles, such as conventional liposomes and adeno-associated viruses (AAVs), suffer from low brain accumulation efficiency (< 0.1% injected dose per gram of tissue), pronounced hepatotoxicity, systemic off-target accumulation, and immunogenicity.

Furthermore, recent genomic and transcriptomic profiling from our team at the **Synapse Institute** has revealed that neurodegenerative progression is strongly driven by hyper-reactive, pro-inflammatory microglial phenotypes (M1-like state) and reactive astrocytes (A1-like state). Restoring brain homeostasis requires non-invasive systems capable of crossing the BBB and selectively targeting dysfunctional neuroglia without inducing systemic immunosuppression.



Figure 1: Conceptual Overview of Proposed Paradigm. The Synapse-Apex nanocarrier platform bypasses endothelial filtration via receptor-mediated transcytosis to achieve targeted microglial polarization and synaptic rescue.

1.2 A.2 Impact on the Field and Public Health

This R01 proposal introduces a paradigm-shifting nanomedicine architecture—the **Synapse-Apex Nanocarrier System**—designed to overcome transcytosis bottlenecks while enabling cell-type specific payload release. Successful completion of this research will:

- **Establish a Universal Transcytosis Platform:** Provide a quantitative benchmark for receptor-mediated endothelial transport across intact human microvascular organoids and in vivo rodent models.
- **Enable Cell-Specific Gene Silencing:** Demonstrate targeted delivery of microRNA-124 mimics directly to hyper-activated microglia, suppressing pro-inflammatory $\text{TNF-}\alpha$, $\text{IL-1}\beta$, and NLRP3 inflammasome signaling.
- **Transform Neurotherapeutic Delivery:** Create a scalable, non-viral delivery toolkit compatible with mRNA, CRISPR-Cas12a complexes, and therapeutic monoclonal antibodies.

1.3 A.3 Scientific Rigor of Prior Research

Prior studies evaluating transferrin receptor (TfR1) antibody-conjugated nanoparticles demonstrated enhanced brain uptake but caused severe reticulocyte depletion and acute receptor down-regulation due to high binding affinity ($K_D < 1$ nM). Work by Dr. Aris Thorne at **Apex Labs** demonstrated that monovalent, medium-affinity ($K_D \approx 50$ –150 nM) cyclic peptides achieve optimal transcytosis efficiency without inducing receptor lysosomal degradation. Building on these preliminary findings, our proposal combines monovalent TfR-binding peptides with microglial-targeting ligand Annexin V to construct a dual-guided targeted nanocarrier.

KEY SIGNIFICANCE HIGHLIGHT

The proposed research addresses the fundamental bottleneck of central nervous system (CNS) drug delivery by combining **low-affinity receptor transcytosis** with **cell-state specific enzymatic activation**, unlocking systemic biotherapeutic treatment for neurodegenerative diseases.